

**BACHELOR OF TECHNOLOGY
ELECTRICAL ENGINEERING
FOURTH YEAR (SEVENTH SEMESTER)
W.E.F. ADMISSION BATCH 2023-24**

Sl. No.	Category	Course Code	Course	L-T-P	Credit	University Marks	Internal Evaluation
Subject (Theory)							
1	PE	EEPE4001	Electric Drives and Control	3-0-0	3	100	50
		EEPE4002	Smart Grid Technology				
		EEPE4003	FACTS and HVDC				
		ECPE3012	Advanced Control Systems				
		-	-				
		-	-				
2	OE	EEPE3005	Renewable Energy Sources	3-0-0	3	100	50
		CIOE4001	Remote Sensing and GIS				
		EOOE4001	Fundamental Communication System				
		MMOE4001	Introduction to Robotics				
		CSPC2003	Object Oriented Programming				
		HSOE4002	Optimization Engineering				
		HSOE4004	Project Management				
		MFPC3005	Computer Aided Design and Manufacturing (CAD / CAM)				
		CSPE3008	Evolutionary Computing				
3	OE	BTOE4001	Bio Imaging	3-0-0	3	100	50
		EEOE4001	Energy Conservation and Auditing				
		CIOE4002	Waste Management				
		EOOE4002	Introduction to Wireless Communication				
		AEOE4001	Introduction to Aerodynamics				
		CSPC2006	Design and Analysis of Algorithms				
		CSOE4001	Design Thinking				
		EVOE4001	Environmental Policy and Legislation				
		HSOE4003	Principles and Applications of Nano Technology				
4	PC(ACC)	HSOE4001	Marketing Management	3-0-0	2	100	50
		DSOE4001	Data Science and Analytics				
		PCAC4001	Applied Data Science				
		PCAC4002	Cloud and Application Development				
		PCAC4003	Cloud Networking				
		PCAC4004	IoT Technologies and Applications				
		PCAC4005	Robotics : Applications				
5	PSI	PCAC4006	Cybersecurity Applications	0-0-6	3	-	100
		PCAC4007	Applied Deep Learning				
Subject (Sessional / Practical)							
5	PSI	EEPS4201	Seminar on SIRE - II	0-0-6	1	-	100
6	PSI	EEPS4202	Project for Product Development - II	0-0-6	3	-	100
		EEPS4203	Internship Project - I				
Total				12-0-12	15	400	400

NB : Subjects registered/completed in earlier semesters as Professional Core or Professional Elective courses shall not be opted for as Open Elective courses.

[Click here to view/download the syllabus of the subjects.](#)

EEPE4001 ELECTRIC DRIVES & CONTROL (3-0-0)

Course objectives:

After completing this course, students will be able to:

1. Understand the dynamics of electrical drives, torque equations, and load characteristics.
2. Analyze starting, braking, and speed control methods of DC, induction, and synchronous motor drives.
3. Study the operation of controlled rectifiers, choppers, and converters used in drive systems.
4. Compare fixed-speed and variable-speed drive techniques for industrial applications.
5. Select appropriate drive and control methods for a given load and operating condition.

Module I (07 Hrs)

Dynamics of Electrical Drives: Introduction, fundamental torque equations, multi-quadrant operations, components of load torque, steady state, and transient stability. Modified characteristics of DC shunt, series motors, and methods of starting. Characteristics of three-phase induction motors and synchronous motors, and starting methods, industrial applications.

Module II (05 Hrs)

Electric Braking: types of braking-regenerative braking, dynamic braking, and plugging, braking of DC motors during lowering of loads and while stopping, electric braking of induction motors, braking of synchronous motors, dynamics of braking.

Module III (05 Hrs)

Introduction to solid state-controlled drives: Controlled rectifier-fed DC drives, single phase and three phase half-controlled and fully controlled rectifier control of DC separately excited motor, rectifier control of DC series motor, chopper control of DC separately excited and DC series motors.

Module III (08 Hrs)

Induction motor Drives: speed control, pole changing, stator voltage control, variable frequency control from voltage sources, closed loop speed control of VSI and CSI fed induction motor drives, Rotor resistance control, slip power recovery, static Scherbius drive, and Kramer drive, variable speed constant frequency generation, starting, braking, and speed control of single-phase induction motors.

Module IV (05 Hrs)

Synchronous motor Drives: Operation from fixed frequency supply, synchronous motor variable speed drives, variable frequency control of multiple synchronous motors, self-controlled synchronous motor drive employing load commutated thyristor inverter, and a cyclo converter

Course Outcomes (CO):

- CO1: Explain the fundamentals of electrical drive dynamics, including torque equations, load torque components, steady-state operation, transient stability, and multi-quadrant operation.
- CO2: Analyze the modified characteristics, starting methods, and operating performance of DC shunt, DC series, induction, and synchronous motors.
- CO3: Compare and evaluate electric braking methods such as regenerative braking, dynamic braking, and plugging for DC, induction, and synchronous motor drives.
- CO4: Analyze controlled rectifier-fed and chopper-fed DC drives, including separately excited and series motor control.
- CO5: Evaluate induction motor speed control techniques such as stator voltage control, pole changing, rotor resistance control, slip power recovery, and variable frequency control from voltage-source and current-source inverters.
- CO6: Explain synchronous motor variable-speed drives, including self-controlled operation, load-commutated thyristor inverter drives, cycloconverter drives, and industrial applications of modern drive systems.

Text Book(s):

1. "Fundamentals of Electrical Drives" by Gopal K. Dubey, second edition, Alpha science int. Ltd., 2001.
2. "A first course on Electrical Drives", by S.K. Pillai, second edition, New Age International (P) Ltd., 2004
3. "Electric Drives" by Nisit K. De and Prasanta K. Sen, Prentice Hall of India Private Limited, 2006.
4. "Electrical Machine Drives and Control- An introduction", by J. Pyrhonen, V. Hrabovcova, R. S. Semken, Wiley, 2016

Reference Book(s):

1. "Control of Electrical Drives", by Werner Leonhard, Second edition, Springer, 1996
2. Mohamed El-Sharkawi, "Fundamentals of Electric Drive," CL- Engineering, 1st Edition, 2000
3. Bimal K. Bose, "Modern Power Electronics and AC Drives," Prentice-Hall, Inc., 2002
4. D. W. Novotny, T. A. Lipo, "Vector Control and Dynamics of AC Drives," Clarendon Press, 1996.
5. R. Krishnan, "Electric Motor Drives - Modelling, Analysis and Control," 1st Edition, Pearson Education, Inc., 2001

EEPE4003 FACTS AND HDVC (3-0-0)

Course Learning Objectives:

This course will enable students to:

- Evaluate the importance of HVDC transmission system, analysis of HVDC converters, and power flow control.
- Compare different control strategies in HVDC systems and HVDC breaker.
- Analyze the working of different FACTS devices and benefits of FACTS over HVDC system.
- To understand the basic operating principle of different type of Shunt and series compensator and their application towards stability and voltage regulation in power system.

Module I 7 Hours

Introduction: Comparison of AC-DC Transmission, Description and application of HVDC transmission, DC System components and their functions, Analysis with gate control with overlap less than 60 degree-operation of inverter, 12-pulse converter, effect of Leakage inductance, power flow in HVDC link

Module II 7 Hours

HVDC Control, Advance Control and Protection: Principles of DC Link control-Converter control characteristics-, Steady state characteristics, ideal and actual, System control, Firing angle control-Current and extinction angle control, DC link power control, Reactive power control and VAR sources, VDCOL- control and protection- DC circuit breakers

Module III 6 Hours

FACTS Concept and General System:

Transmission interconnections, Flow of power in AC system, Power flow and dynamic stability considerations of a transmission interconnection, Relative importance of controllable parameters, Basic types of FACTS controllers, Benefits from FACTS Technology, In-perspective: HVDC or FACTS

Module IV 6 Hours

Static Shunt and Series Compensators: Static Shunt Compensators - SVC and STATCOM - operation and control of TSC, TCR, STATCOM - Compensator Control - Comparison between SVC and STATCOM - STATCOM for transient and dynamic stability enhancement.

Static Series Compensation - TSSC, TCSC and SSSC - operation and control - external system control for series compensators - SSR and its damping - static voltage and phase angle regulators.

Module V 4 Hours

UPFC and IPFC: The Unified Power Flow Controller - operation, comparison with other FACTS devices - control of P and Q - dynamic performance - Special Purpose FACTS Controllers - Interline Power Flow Controller - operation and control

Course Outcomes (CO): On completion of this course, students are able to:

- CO1. Compare AC and DC transmission systems and explain the structure, components, and functions of an HVDC transmission scheme
- CO2. Analyse the operation of converters with gate control, including the effect of leakage inductance and other advance power control
- CO3. Explain the concept of FACTS, the flow of power in AC transmission systems, Classification of the basic types of FACTS controllers
- CO4. Understand and Analyse different shunt and series thyristor based FACTS controller like SVC, TCSC
- CO5. Understand the working principle VSC based FACTS technology like STATCOM, SSSC, UPFC, IPFC

Text Book(s):

1. Narain G. Hingorani and L. Gyugyi, 'Understanding FACTS', IEEE Press.
2. HVDC Transmission, S. Kamakshaiah, V. Kamaraju, The Mc- Graw Hill Companies.
3. HVDC Power Transmission Systems: Technology and system Interactions, K. R. Padiyar, New Age International (P) Limited.

Reference Book(s):

1. J. Arrillaga, Peter Pregrinu, High Voltage Direct Current Transmission.
2. HVDC and FACTS Controllers Applications of Static Converters in Power Systems, Vijay K. Sood, Kluwer Academic Publishers.

EEPE3005 RENEWABLE ENERGY SOURCES (3-0-0)

Course Learning Objectives: This course will enable to understand the students the working, characteristics, and types of renewable energy sources.

- To make the students analyze the energy conversion mechanisms of the renewable energy sources and hybrid systems.
- To enable students to integrate and implement renewable energy systems in power grids.
- To make students optimize the technical, economic, and environmental performance of renewable energy technologies.

Module I (10 Hours)

Introduction: Types of energy sources (Conventional and non-conventional) and its Impacts, Worldwide potentials of these sources, Energy consumption patterns, Distributed generation. Renewable Sources: Renewable energy sources and features, Solar Energy: Solar thermal Systems, Types of collectors, Collection systems, efficiency calculations, applications. Solar photovoltaic system-Operating principle, Characteristics of PV systems, Equivalent circuit, Cell, Module, Array, Series and parallel connections, Cell mismatching, Maximum power point tracking, Standalone and grid interactive systems, Peak power operation, Applications of solar PV, Modeling of PV cell.

Module II (8 Hours)

Wind Energy: Wind energy, Wind energy conversion, Types of wind converters, System components, Aerodynamic operation of wind turbines, Maximum power extracted from wind, Wind speed and power characteristics, Choice of generators, turbine rating, electrical load matching, Variable speed operation, single and double output systems, DFIG concepts, Reactive power compensation, Control systems, System design features, Stand alone and grid connected operation.

Module III (6 Hours)

Small Hydro Systems: Layout of a Micro-Hydro scheme, Water turbines: Classification and characteristics, Present status.

Bio-Mass: Principles of biomass conversion, Combustion and fermentation, Anaerobic digestion, Classification of Biogas plants, Present status

Tidal power: Tidal phenomena, Tidal power development and tide mills, Ocean tidal energy conversion schemes, Present status

Module IV (7 Hours)

Energy storage Technologies: Energy storage, Need of energy storage, Specifications, Types of storage. Battery management. Fuel Cell energy storage systems. Ultra Capacitors.

Hybrid Systems: Need for hybrid systems, Range and type of hybrid systems, Hybrid Micro hydel-PV, PV-Wind systems, Electric and hybrid electric vehicles.

Course Outcomes (CO): On completion of this course, students are able to:

- CO1. Discuss the concepts of conventional and non-conventional energy sources along with their environment impacts, and the importance of solar power systems
- CO2. Evaluate the operation and control of wind energy conversion systems, turbine designs, DFIG operation, and reactive power compensation.
- CO3. Describe the principles and technologies of small hydro, biomass, and tidal energy systems and assess their current status and applications.
- CO4. Assess different energy storage systems and hybrid renewable configurations, including battery management, fuel cells, and hybrid electric vehicles.

Text Book(s):

1. B.H.Khan, "Non-Conventional Energy Resources", Tata McGrawHill, 2009.

2. S. N. Bhadra, D. Kastha, S. Banerjee, "Wind Electrical Systems", Oxford University Press, 2005.
3. Renewable energy sources and emerging technologies -D.P. Kothari, K.C. Singal, and R. Ranjan, PHI Learning Pvt. Ltd.

Reference Book(s):

1. Modern Electric Vehicle Technology by C.C. Chan and K.T. Chau.
2. O. I. Elgerd, Electric Energy Systems Theory. New York, NY, USA: McGraw-Hill Education, 1995.
3. B. M. Weedy, B. J. Cory, N. Jenkins, J. Ekanayake, and G. Strbac, Electric Power Systems, 5th ed. Hoboken, NJ, USA: Wiley, 2012.
4. R. D. Bergen and V. Vittal, Power System Analysis. Upper Saddle River, NJ, USA: Pearson Education Inc., 1999.

EEOE4001 ENERGY CONSERVATION & AUDITING (3-0-0)

Course Objectives:

This course provides comprehensive knowledge of energy conservation and auditing techniques, covering energy management principles, industrial audits, electrical system efficiency (motors, HVAC, lighting), power factor improvement, and economic analysis methods (payback, NPV). Students will learn to identify energy-saving opportunities and evaluate conservation projects using practical tools and methodologies.

Module-I: (06 Hrs.)

Basic principles of Energy audit: Energy audit-definitions, concept, types of audit, energy index, cost index, pie charts, Sankey diagrams, load profiles, Energy conservation schemes-Energy audit of industries-energy saving potential, energy audit of process industry, thermal power station, building energy audit

Module-II: (05 Hrs.)

Energy management: Principles of energy management, organizing energy management program, initiating, planning, controlling, promoting, monitoring, reporting.

Module-III: (08 Hrs.)

EB billing: HT and LT supply, transformers, electric motors- motor efficiency computation, energy efficient motors, pumps, fans, blowers, compressed air systems, refrigeration and air conditioning systems, cooling towers, electric heaters (space and liquid), DG-sets, illuminating devices, power factor improvement, and harmonics.

Module-IV: (06 Hrs.)

Power factor improvement, Lighting: Power factor-methods of improvement, location of capacitors, pf with nonlinear loads, effect of harmonics on p.f., p.f. motor controllers-good lighting system design and practice, lighting control, lighting energy audit

Module-V: (05 Hrs.)

Economic aspects and analysis: Economic analysis- Depreciation methods, time value of money, rate of return, present worth method, replacement analysis, life cycle costing analysis-Energy efficient motors
Computation of Economic aspects: Calculation of simple payback method, net present worth method-Power factor correction, lighting-Applications of life cycle costing analysis, return on investment.

Course Outcomes:

On completion of this course, students are able to:

- CO1: Understand and explain the basic principles, concepts, and methodologies used in energy audits and identify opportunities for energy conservation.
- CO2: Apply energy management principles to plan and organize an effective energy management program in an industrial or commercial setup.
- CO3: Analyze energy consumption patterns in electrical systems such as motors, pumps, fans, and HVAC systems and suggest efficiency improvements.
- CO4: Evaluate power factor correction and lighting systems, and perform energy audits to enhance energy efficiency and quality.
- CO5: Perform economic analysis of energy conservation projects using techniques such as payback period, NPV, and life cycle cost analysis.

Text Book(s):

1. Paul o' Callaghan, Energy management, Mcgraw Hill Book company, 1998
2. W.R.Murphy and G.McKay, Energy Management, Butterworth Heinemann, 1978
3. John C. Andreas, Energy efficient electric motors, Marcel Dekker Inc Ltd, 1995
4. S.C.Tripathy, Electric Energy Utilization & Conservation, Tata Mcgraw Hill, 1991

Reference Book(s):

1. W.C.Turner, Energy Management Handbook, Fairmont Press, 1993
2. Albert Thumann, William J. Younger, Handbook of Energy Audits, Terry Niehus, 2009

ACC: ACC 3

ACC Bucket : Python & Machine Learning - 3

Course: Applied Deep Learning

OVERALL COURSE OBJECTIVES: The objective of this course is to synthesize comprehensive deep learning knowledge, cultivate capabilities to configure and optimize neural networks, refine machine learning troubleshooting strategies, and apply advanced concepts, all while actively contributing to the development of leading-edge AI technology.

LEARNING OUTCOMES: On successful completion of the course the students shall be able to:

1. Construct, train, and implement fully connected deep neural networks and apply them to various applications.
2. Evaluate and employ standard neural network techniques such as initialization, L2 and dropout regularization, hyperparameter tuning, batch normalization, and gradient checking.
3. Implement a variety of optimization algorithms, including mini-batch gradient descent, Momentum, RMSprop and Adam, and check for their convergence.
4. Implement a neural network in TensorFlow.
5. Diagnose errors in a machine learning system and strategize solutions for reducing these errors.
6. Apply advanced machine learning concepts such as end-to-end learning, transfer learning, and multi-task learning.

COURSE CONTENT:

Module 1: [Neural Networks and Deep Learning](#) [25 Hours]

In this foundational course of the Deep Learning Specialization, you will delve into neural networks and deep learning. The course aims to familiarize you with significant technological trends propelling deep learning, enabling you to build, train, and apply fully connected deep neural networks. It will also guide you to implement efficient (vectorized) neural networks, identify key parameters of a neural network's architecture, and apply deep learning to your applications. This specialization forms a cornerstone for understanding the potential, challenges, and implications of deep learning. It creates an avenue for you to gain the proficiency to employ machine learning in your work, elevate your technical career, and make significant strides in the AI landscape.

Sub-Topic

Supervised Learning with Neural Networks
Binary Classification
Derivatives with a Computation Graph
Vectorizing Logistic Regression
Shallow Neural Networks
Building Blocks of Deep Neural Networks

Formative Assessments:

4 quizzes and 4 graded coding/lab assignments.

Module 2: [Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization](#) [24 Hours]

The second course in the Deep Learning Specialization delves into understanding the mechanics behind deep learning performance and consistently generating good results. Upon completion, you will master best practices for training, developing test sets, and analyzing bias/variance for deep learning applications. Additionally, you will acquire skills in implementing standard neural network techniques such as initialization, L2 and dropout regularization, hyperparameter tuning, batch normalization, gradient checking, and a variety of optimization algorithms. The course also covers implementing a neural network in TensorFlow. This specialization is foundational and prepares you for participation in advanced AI technology development, enabling you to apply machine learning practically, accelerate your technical career, and make significant strides in the AI domain.

Sub-Topic

Practical Aspects of Deep Learning
Understanding Exponentially Weighted Averages
Weight Initialization for Deep Networks
The Problem of Local Optima
Optimization Algorithms
Hyperparameter Tuning, Batch Normalization and Programming Frameworks

Formative Assessments:

3 quizzes and 5 coding/lab assignments

Module 3: [Structuring Machine Learning Projects](#) [7 Hours]

The third course of the Deep Learning Specialization aims to equip you with the ability to successfully execute a machine-learning project, cultivating your decision-making skills as a project leader. You will acquire the ability to diagnose errors in a machine learning system, prioritize error-reducing strategies, comprehend complex ML settings, and apply end-to-end learning, transfer learning, and multi-task learning. This course is suitable for learners with basic machine learning knowledge and offers industry experience drawn from Andrew Ng's extensive deep learning product development. It's perfect for those aspiring to become technical leaders in AI teams. The Deep Learning Specialization as a whole provides a sturdy foundation on which you can explore the capabilities, challenges, consequences of deep learning, and apply machine learning to your work, thus enhancing your technical career and consolidating your steps in the AI world.

Sub-Topic

Why ML Strategy
Understanding Human-level Performance
Surpassing Human-level Performance
Transfer Learning
Multi-task Learning
What is End-to-end Deep Learning?

Formative Assessments:

2 quizzes and 5 coding/lab assignments

ASSESSMENT:

For summative assessments, Coursera will provide question banks for which exams can be conducted on the Coursera platform or the faculty will create their own assessments.

Note: If a Course or Specialization becomes unavailable prior to the end of the Term, Coursera may replace such Course or Specialization with a reasonable alternative Course or Specialization.